1. Introduction

1.1. Brief overview of the selected SDG use case

This project addresses the United Nations' Sustainable Development Goal 3 (SDG 3): Good Health and Well-being, which aims to ensure healthy lives and promote well-being for all at all ages. A critical, yet often invisible, threat to this goal is urban air pollution. Poor air quality is a major environmental risk to public health, contributing to a wide range of respiratory and cardiovascular diseases.

To address this challenge, this project implements a data engineering solution focused on the real-time analysis of air quality. By leveraging a comprehensive dataset of atmospheric pollutant concentrations, our use case is to build a robust data pipeline that can ingest, process, and analyze air quality data. The goal is to transform raw sensor readings into actionable insights that can help public health officials, urban planners, and the general public make informed decisions to mitigate the health impacts of air pollution.

1.2. Motivation and real-world relevance

The motivation for this project is grounded in the pressing public health concerns associated with air pollution in rapidly developing urban areas. In Malaysia, haze and high concentrations of pollutants like carbon monoxide (CO), nitrogen dioxide (NO_2), and benzene are significant issues that directly impact the well-being of the population. These pollutants can trigger or worsen conditions such as asthma, bronchitis, and other chronic respiratory illnesses, particularly affecting vulnerable groups like children and the elderly.

The real-world relevance of this pipeline has 2 different perspectives. Firstly, the historical analysis of the data can inform smarter urban planning and environmental policy. By identifying patterns such as the most polluted times of day or days of the week, the data can provide evidence to support initiatives like promoting public transport or regulating industrial emissions. Finally, by making this data accessible and understandable through a dashboard, it empowers citizens with the knowledge to make personal decisions to reduce their exposure to harmful pollutants.

1.3. Objectives of the pipeline

a) **To Enable Proactive Public Health Interventions**: The primary objective is to transform raw, real-time sensor data into actionable health intelligence. By identifying high-risk pollution events and analyzing patterns such as peak pollution hours, the pipeline aims to provide public health officials with the evidence needed to issue timely health advisories, particularly for vulnerable populations, thereby reducing exposure and preventing acute respiratory incidents.

b) **To Empower Citizen Awareness and Action**: A key goal is to make complex environmental data accessible and understandable to the public. By storing the processed data in a query-friendly format and visualizing it on a dashboard, the pipeline aims to empower individuals and communities with the knowledge to understand their local air quality, enabling them to make informed decisions to reduce their personal exposure and advocate for a healthier environment.

c) **To Identify and Quantify Weekly Patterns in Air Pollution:** By aggregating data for each day of the week, the pipeline can move beyond individual readings to see the bigger picture of a typical week's rhythm such as how the workdays and the weekends impacts the air quality and, consequently, public health. This can link to human activity cycles and provide evidence for policy-making.

2. Task 1: Raw Data Streaming

Use Case:	Real-time Air Pollution Data Collection
Raw Data Source:	Replaying historical data as if it were streaming in real time ~/de-prj/de_assignment/src/data/AirQuality.csv
Kafka Topic:	“air_quality_stream”
Output Path:	"hdfs://localhost:9000/user/air_quality/raw"

List of Python (`.py`) files:

Python file(s)	Author
task1a_consumer.py	Lim Jun Jie
task1b_producer.py	Lim Jun Jie
spark_manager.py	Desmond Oon Kai Quan

3. Task 2: Data Processing

List of Python (`.py`) files:

Python file(s)	Author
task2a_process_data.py	Desmond Oon Kai Quan

spark_manager.py	Desmond Oon Kai Quan

4. Task 3: Document Database

4.1. MongoDB Data Model Design

Sensor_readings

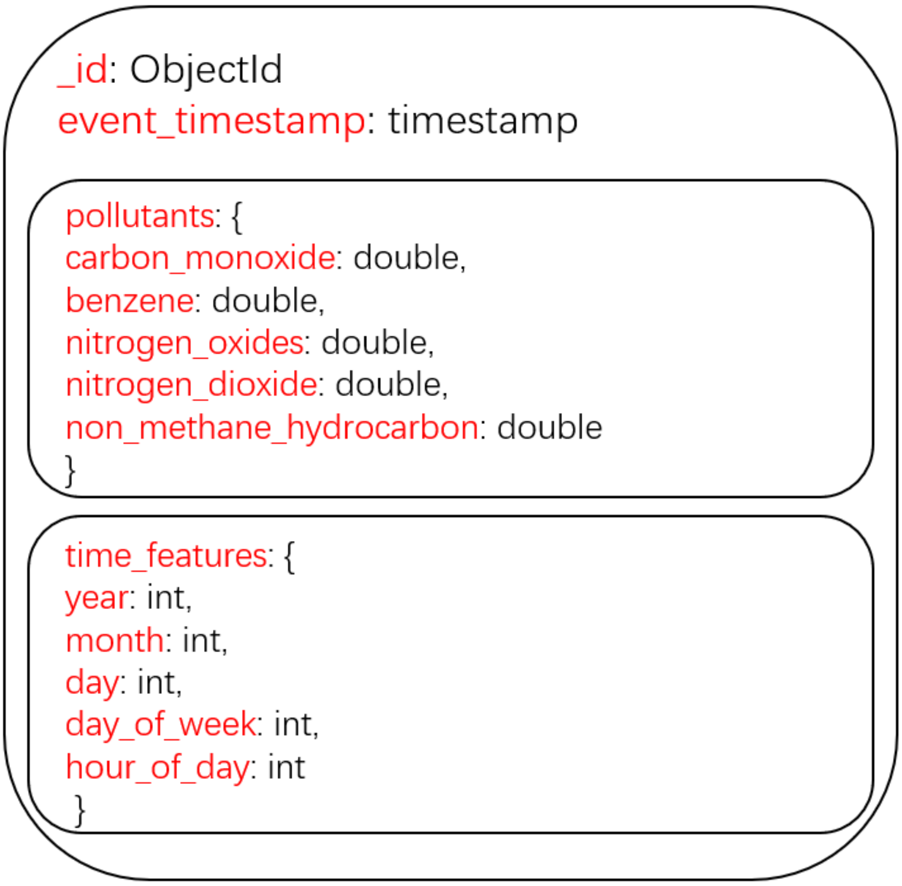

Figure 1: MongoDB Data Model Design

4.2. Diagram with Example of Values in MongoDB

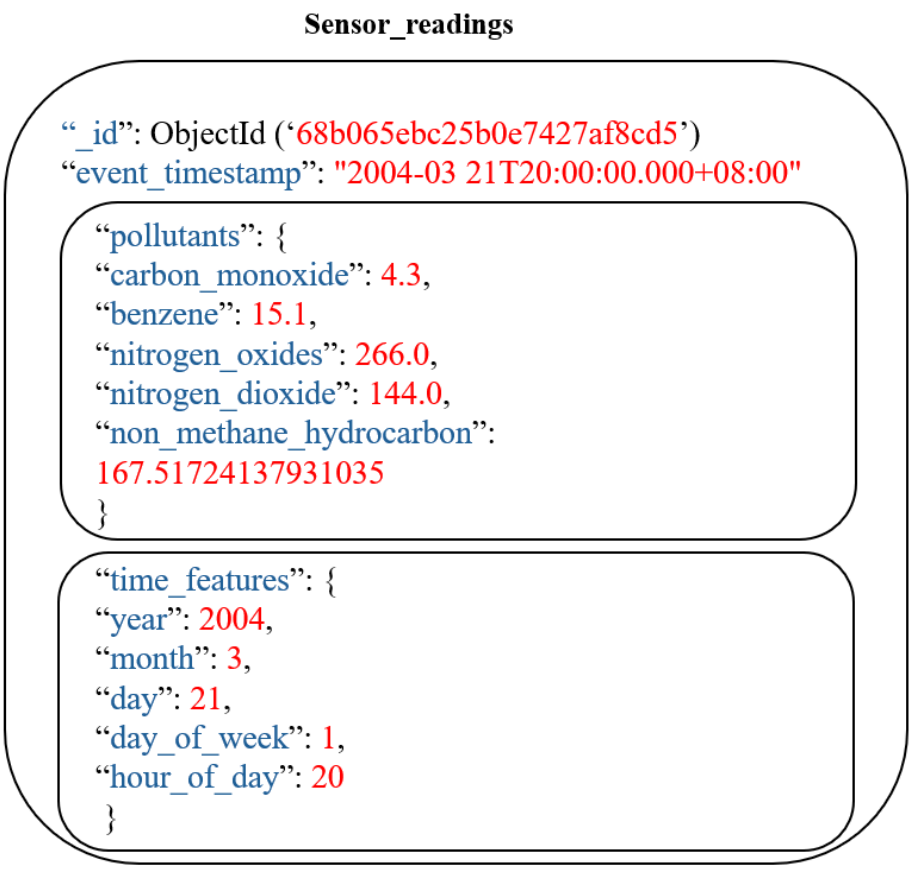

Figure 2: Diagram with Example Values in MongoDB

4.3. List of Python (`.py`) files:

Python file(s)	Author
task3a_load_to_mongo.py	Phung Jian Thong
task3b_mongo_query.py	Phung Jian Thong
task3c_mongo_visualisation_seaborn.py	Phung Jian Thong
mongo_connector.py	Phung Jian Thong
spark_manager.py	Desmond Oon Kai Quan

5. Graph Database

5.1. Graph Model Design

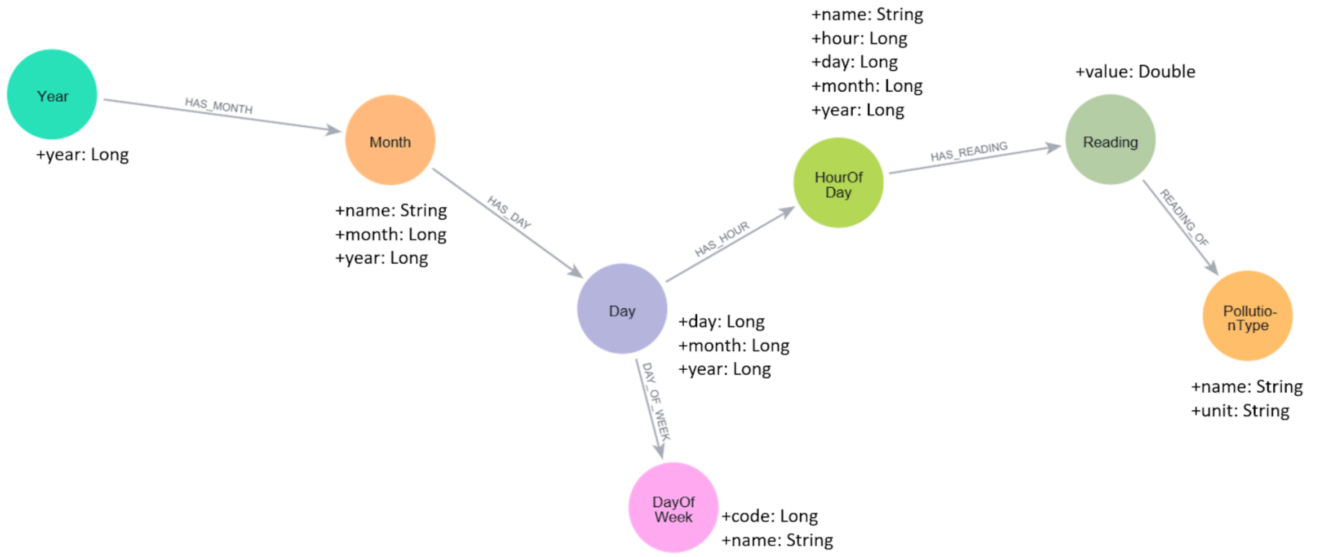

Figure 3: Graph Data Model Design

5.2. Diagram with Example of Values in Neo4j

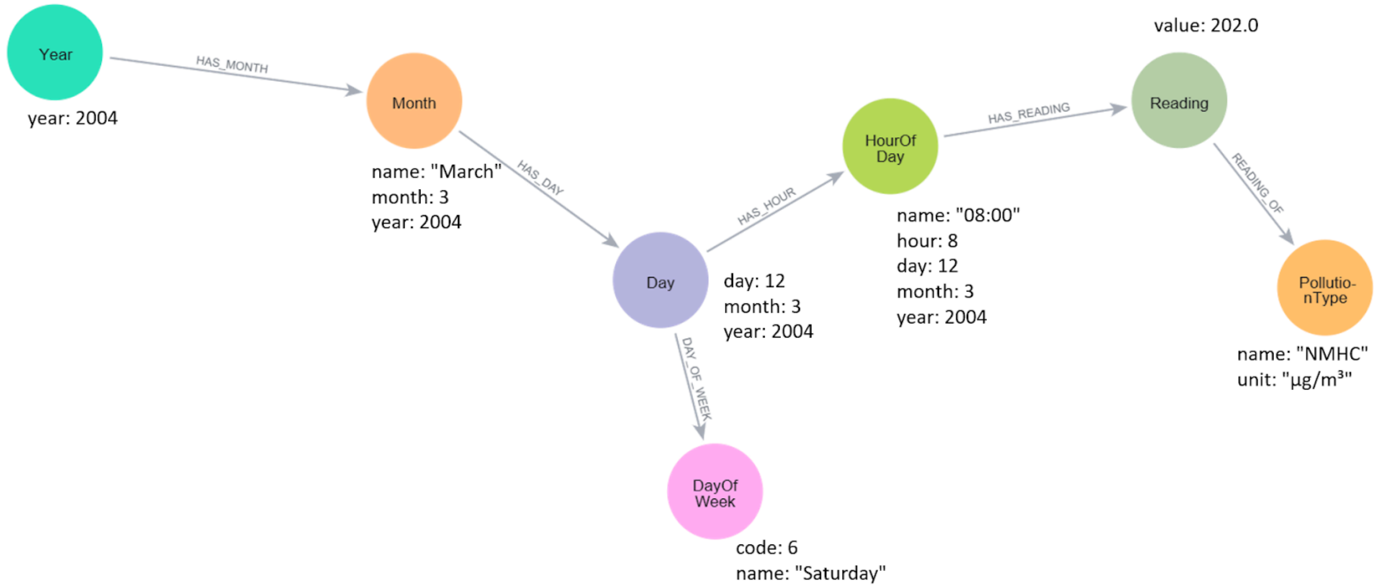

Figure 4: Diagram with Example of Values in Neo4j

5.3. List of Python (`.py`) files:

Python file(s)	Author
task4a_load_to_neo4j.py	Seow Zhan Hou
task4b_neo4j_query.py	Seow Zhan Hou
task4c_neo4j_visualisation.py	Seow Zhan Hou
neo4j_connector.py	Seow Zhan Hou
spark_manager.py	Desmond Oon Kai Quan

6. Spark Structured Streaming

6.1. Structured Streaming Workflow Diagram

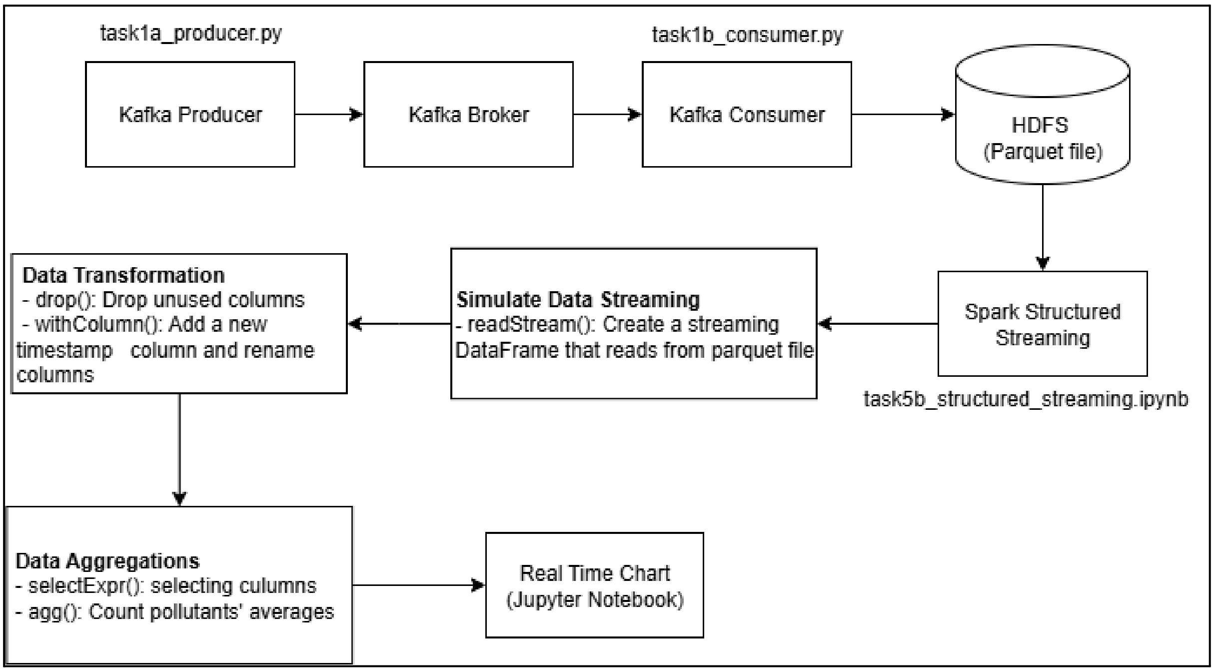

Figure 5 : Structured Streaming Workflow Diagram

6.2. List of Python (`.py`) files:

Python file(s)	Author
task5a_structured_streaming.py	Lee Kay Yang
spark_manager.py	Desmond Oon Kai Quan